FMCG REVIEW ANALYTICS

Product Sentiment Analysis

Classifying customer reviews with TF-IDF and Logistic Regression

ILLUSTRATIVE EXAMPLE

Love the scent, but the bottle leaks. Not worth the price.

Predicted sentiment **Negative**

TF-IDF | Logistic Regression | Confusion matrix

01 · PROBLEM

The problem

Automated product sentiment analysis often struggles with negation blindness and mixed-sentiment detailed reviews.

Bag-of-words view of 'This soap is not good'

This | soap | is | not | **good**

Predicted: Positive

Negation blindness

'Not good' is scored by the word 'good' alone, so negative reviews flip positive.

Mixed-sentiment reviews

Long reviews praise one feature and criticise another, so one label hides the mix.

Unaudited length gaps

Accuracy can differ between short and long reviews, and the gap is rarely audited.

02 · OBJECTIVE

The objective

Develop and benchmark an interpretable sentiment pipeline that compares TF-IDF with Transformers.

01

Compare

Benchmark TF-IDF against Transformer models.

02

Resolve

Reduce negation errors such as 'not good'.

03

Audit

Measure gaps between short and long reviews.

04

Deploy

Ship a real-time, interactive showcase.

03 · SOLUTION

Solution: TF-IDF + Logistic Regression

A transparent pipeline that turns review text into weighted word features, then into a sentiment label.



03 · SOLUTION

Dataset: 3,500 FMCG reviews

3,500
customer reviews

8
columns per review

Source: Kaggle, FMCG Customer Sentiments

Column	Role
review_text	Input (X): the review itself
sentiment_label	Target (y): sentiment class
rating	Star rating given
review_length	Length of the review
brand	Brand name
product_category	Broad product category
product_type	Specific product type
review_id	Unique review ID

03 · SOLUTION

TF-IDF: turning words into numbers

TERM FREQUENCY (TF)

$$TF(t, d) = \text{count of } t \text{ in } d \;/\; \text{total terms in } d$$

INVERSE DOCUMENT FREQUENCY (IDF)

$$IDF(t) = \log\left(N \;/\; (1 + df(t)) \right)$$

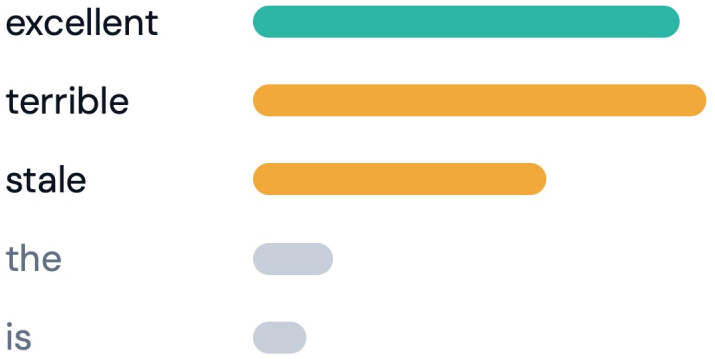
N = total reviews, df(t) = reviews containing t

TF-IDF SCORE

$$TF\text{-}IDF(t, d) = TF(t, d) \times IDF(t)$$

Which words carry weight

Illustrative: a longer bar means a higher TF-IDF weight



03 · SOLUTION

Logistic Regression: from weights to a label

1 · SCORE PER CLASS (X = TF-IDF VECTOR)

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z_k = w_k · x + b_k
```

2 · PROBABILITY (SOFTMAX)

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P(y = k | x) = e^(z_k) / sum_j e^(z_j)
```

3 · PREDICTION

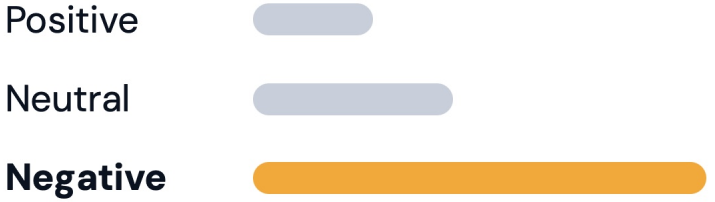
```
label = class with the highest P
```

TRAINING

```
loss = cross-entropy + L2 penalty
```

From probabilities to a label

Illustrative output for one review



Predicted: Negative

03 · SOLUTION

How the solution answers the problem

Negation

Word pairs such as 'not good' become their own features, so the model can learn they mean negative.

Mixed sentiment

Class probabilities flag borderline reviews, and the confusion matrix shows where mixed reviews are misread.

Review length

The review_length column lets us compare accuracy on short and long reviews.

Explainability

Each coefficient belongs to a word, so the words behind a prediction can be shown.

03 · SOLUTION

Evaluate and explore in the live demo

- See the confusion matrix of actual vs predicted sentiment
- Read accuracy, precision, recall and F1-score
- Calculate TF-IDF scores for any review text
- Try your own product review

LIVE DEMO

Try your own review

product-sentiment-analysis-ashen.vercel.app